Experiment ‘p_6_1_c_process_array_reverse_order_array’ Results

June 5, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_1_c_process_array_reverse_order_array
Natural language statement: null
Method signature: p_6_1_c_process_array_reverse_order_array(inputs: array<int>) returns (reversed: array<int>)

Ensures

- `reversed.Length == inputs.Length`
- `forall i :: 0 <= i < reversed.Length ==> reversed[i] == inputs[inputs.Length - 1 - i]`

Requires

- `inputs.Length == 10`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_6_1_c_process_array_reverse_order_array(inputs: array<int>) returns (reversed: array<
    ↪ int>)

The method should respect the following contract:

requires inputs.Length == 10, ensures reversed.Length == inputs.Length, ensures forall i
    ↪ :: 0 <= i < reversed.Length ==> reversed[i] == inputs[inputs.Length - 1 - i]

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnNegdWLzE2v8L4Pr54D7qqCuSwFg
Model: gpt-5.5-2026-04-23
Created at: 1780661374
Finish reason: stop
Usage: 324 tokens in, and 635 tokens out

```
{
  reversed := new int[inputs.Length];
  var i := 0;
  while i < inputs.Length
    invariant reversed.Length == inputs.Length
    invariant 0 <= i <= inputs.Length
    invariant forall j :: 0 <= j < i ==> reversed[j] == inputs[inputs.Length - 1 - j]
  {
    reversed[i] := inputs[inputs.Length - 1 - i];
    i := i + 1;
  }
}
```

Final Program

```
method p_6_1_c_process_array_reverse_order_array(inputs: array<int>) returns (reversed:
    ↪ array<int>)
```

```
      requires inputs.Length == 10
      ensures reversed.Length == inputs.Length
      ensures forall i :: 0 <= i < reversed.Length ==> reversed[i] == inputs[inputs.
          ↪ Length - 1 - i]
{
  reversed := new int[inputs.Length];
  var i := 0;
  while i < inputs.Length
    invariant reversed.Length == inputs.Length
    invariant 0 <= i <= inputs.Length
    invariant forall j :: 0 <= j < i ==> reversed[j] == inputs[inputs.Length - 1 - j]
  {
    reversed[i] := inputs[inputs.Length - 1 - i];
    i := i + 1;
  }
}
```

Total Token Usage

Input tokens: 324
Output tokens: 635
Reasoning tokens: 512
Sum of 'total tokens': 959

Experiment Timings

Overall Experiment started at 1780661373798, ended at 1780661408924, lasting 35126ms (35.13 seconds)
Iteration #1 started at 1780661373799, ended at 1780661408924, lasting 35125ms (35.13 seconds)